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Using Place-Based Jobs Policies to Help Distressed Communities

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This article focuses on place-based jobs policies: policies that provide assistance to individual businesses to encourage job growth in a particular local labor market, such as a metropolitan area. When run by state and local governments, place-based jobs policies are commonly called economic development policies. Such business assistance includes business tax incentives, cash grants to business, and special public services to business such as customized job training, manufacturing extension services, small business development centers, business incubators, business parks, and access roads.

Place-based jobs policies currently cost around $60 billion annually (as discussed in the next section). Over the past 30 years, the typical state and local incentive offer, as a percent of the value-added of the business receiving the incentive, has tripled in size. Do current policies make sense? How should they be reformed? Can reforms be carried out by state and local governments, or is federal intervention needed? These are some of the questions I will address.

One rationale for place-based jobs policies is that local labor markets have large and persistent differences in job availability. A commonly-used definition of local labor markets is commuting zones, which are groups of counties which contain most commuting flows; commuting zones are similar to metro areas, but include all US counties. Consider the distribution across commuting zones of the "prime-age employment rate"—that is, the employment-to-population ratio for 25–54 year-olds, as provided by the 2018 American Community Survey (Ruggles et al. 2020). Commuting zones at the median of the population distribution had a prime age employment rate of 80.9 percent, compared with 75.5 percent at the 10th percentile

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and 84.5 percent for the 90th percentile—a 90/10 gap of 9.0 percentage points. Moreover, spatial differences in employment rates persist for decades (Amior and Manning 2018; Austin, Glaeser, and Summers 2018).

These low employment rates in certain local labor markets have large costs both for the individual nonemployed and for the broader society. For the individual, nonemployment not only reduces earnings but leads to poorer mental health (Diette et al. 2018). Local job losses lead to increases in alcohol and drug use and to opioid deaths (Autor, Dorn, and Hanson 2018). Social costs of local labor market distress include the following five items: First, local labor market distress leads to increases in property crime (Pierce and Schott 2017). Second, such distress leads to increases in single-mother families (Autor, Dorn, and Hanson 2018). Third, lower parental income is associated with children doing worse in educational attainment and adult income (Bastian and Michelmore 2018; Stuart 2017). Fourth, declines in local employment rates cause fiscal problems—for example, increased welfare benefits such as disability (Charles, Hurst, and Schwartz 2018). Fifth, an individual's life satisfaction is negatively affected by both their own unemployment and the area's unemployment (Helliwell and Huang 2014).

In the past, researchers such as sociologist William Julius Wilson (1996) have talked about the social problems that develop when work disappears in inner cities. Today, it is widely recognized that local joblessness and its accompanying social problems have spread to many smaller cities and rural areas, as documented by writers such as J.D. Vance in *Hillbilly Elegy* (2016).

To alleviate the costs of local labor market distress, we might want to encourage businesses to create jobs in these distressed areas. The next section describes current place-based jobs policies.[1] The bulk of the article then discusses market failures that might justify place-based jobs policies. This analysis leads to suggested reforms, which could be implemented by state and local governments but might also be promoted by the right type of federal intervention.

What are Place-Based Jobs Policies?

Local-area job growth is potentially affected by many government actions. Here, I focus on state and local government interventions aimed at increasing job growth in a state or in a local labor market, often referred to as "economic development" policies. These policies include tax breaks or cash that are awarded in a discretionary fashion to individual businesses. They also include services customized to individual businesses, like customized job training for a specific firm or advice for firms provided through manufacturing extension services and small business development centers. State and local economic development policies can also include infrastructure investments and land development practices.

[1] This article does not discuss another type of place-based policy: community development policies, which seek to improve places much smaller than a metro area, such as a census tract or other small neighborhood.

Table 1
Resources Devoted to Place-Based Jobs Policies in the United States

Current programs	
Policy/program	*Annual dollars (in billions)*
State and local business tax incentives and other cash incentives	47.1
Non-financial incentives: customized training programs	0.6
Other state economic development programs	2.8
Subtotal, state/local programs	50.6
Manufacturing extension (federal/state/fees)	0.4
Economic Development Administration (EDA)	0.3
Economic development portion of HUD's Community Development Block Grants	1.1
Small Business Administration	0.8
Other economic development programs in USDA, HUD, Commerce	2.0
Subtotal mostly federal spending	4.7
Opportunity Zones tax credits	1.6
New markets tax credit	1.5
Other tax expenditures that might promote local economic development	2.4
Subtotal, federal tax expenditures	5.4
Total of federal programs and tax expenditures	10.1
Total of all levels of government	60.7
Past programs	
Appalachian Regional Commission (peak annual spending 1966–1975)	1.7
Tennessee Valley Authority (peak annual spending 1950–1955)	1.5

Note: All figures in 2019 dollars. State/local cash incentives are 2015 data from Bartik (2017a). Customized job training spending from Hollenbeck (2013). Other state economic development programs from State Economic Development Expenditure Database (Council for Community and Economic Research 2018). To avoid double-counting, I subtract out workforce preparation, strategic business attraction, and business assistance, and subtract out the half of state economic development spending which is federally-financed (Council for Community and Economic Research 2017). This spending includes: tourism, film promotion, other special industry promotion, high-tech programs, business finance, entrepreneurial assistance, minority business development, community assistance, business recruitment, and trade promotion. Manufacturing extension from T. Bartik (2018b). EDA, HUD, and SBA from FY 2017 US federal budget. For CDBG, assume one-third of funds are for economic development. Other economic development spending from GAO (2012b). Opportunity Zones from Kimbo and Phillips (2018). New markets tax credit costs from US Department of Treasury (2016). Other tax expenditures from GAO (2012a). ARC figures from Jaworski and Kitchens (2019). TVA figures from Kline and Moretti (2013).

With this focus, Table 1 summarizes current place-based jobs policies. By far the largest place-based jobs policies are state and local financial incentives to business provided via tax breaks or cash. Such financial incentives were $47.1 billion (in 2019 dollars) in 2015, almost four-fifths of the annual total for all place-based jobs policies of $60.7 billion.[2] State and local governments also provide non-financial

[2]Some estimates of incentives are larger: Thomas (2011) calculates $73 billion and Story (2012), $101 billion (in 2019 dollars). However, Thomas includes tax breaks that are not aimed at job growth. Story includes exemptions of business inputs from sales taxes, generally regarded as a desirable tax feature. Slattery and Zidar (2020) get lower estimates at $22 billion (2019 dollars) for state incentives only, based on state budgets and tax expenditure reports.

incentives through customized job training programs under which community colleges provide free training to firms locating or expanding a facility—but these non-financial incentives are only $0.6 billion. I now focus on describing how state and local incentives are designed and distributed. This description is based on Bartik (2017a), where I use a hypothetical firm model and construct a database which calculates incentives paid over 20 years for a new facility (started up in years 1990-2015) in each of 45 industries and 33 states. The 33 states make up 92 percent of US GDP; the 45 industries make up 91 percent of US compensation.

To put the annual incentives of $47.7 billion in perspective, this total is almost identical to the $47 billion in state corporate income taxes (US Census Bureau 2020), which sounds like a lot. On the other side, $48 billion is less than 3 percent of state and local own-source tax revenue. As we will discuss, state and local governments target incentives at businesses selling outside the state—the "export-base" or "tradable goods" sector. For firms in export-base industries, typical incentives offset 30 percent of a business's state and local taxes. But the typical incentive package equals only 1.4 percent of the business's value-added, or 3.1 percent of wages for such businesses.[3]

Although most incentives are provided by state governments, 27 percent come from local governments through property tax abatements. The largest incentive is job creation tax credits, typically defined as a percentage of the increased wage bill at the new or expanded business, which make up 45 percent of total incentives. Many job creation tax credits are refundable—that is, they can exceed the firm's state corporate income tax liabilities. Some states allow a new or expanding firm to keep the workers' state income tax withholdings associated with the new jobs. More minor roles are played by investment tax credits (14 percent) and research and development (R&D) tax credits (9 percent). Customized job training incentives are less than 5 percent of total incentives.

The share of incentives received by large firms is disproportionate. Firms with over 100 employees get more than 90 percent of incentives (Chatterji 2018), while firms of this size represent only 66 percent of private jobs (based on the Longitudinal Business Database for 2016, from US Census Bureau 2018). Incentives particularly go to the very largest firms. Slattery and Zidar (2020) find that for new establishments with more than 1,000 employees, over 36 percent receive incentives, versus less than 10 percent for establishments between 500 and 999 employees, less than 2 percent for establishments between 250 and 499 employees, and even lower percentages for smaller establishments.

As mentioned, incentives are usually targeted at "export-base industries"—industries that sell their goods and services outside the state. In 2015, in the average state, the present value of incentives, as a percentage of the present value of value-added, was 1.42 percent for export-base industries versus 0.16 percent for non-export-base industries (Bartik 2017a, Table 5).

[3]This is the present value of the incentive package as a percent of the present value of value-added or wages as detailed in Bartik (2017a).

Within a state's export-base industries, incentives are not targeted by whether an industry is high-tech. As shown later, high-tech industries offer greater benefits to a local economy. Of 31 industries analyzed in Bartik (2017, Table 7), the most high-tech, as measured by high ratios of research and development spending to value-added, are chemicals manufacturing and computer manufacturing. Average incentives for chemicals and computers are 1.41 percent and 1.74 percent, ranking them twenty-fourth and tenth among the 31 industries.

In addition, incentives are not tightly tied to whether a state is short of jobs. The circles in Figure 1 each represent a state, with the circles proportional to population size. The horizontal axis shows the prime-age employment rate, while the vertical axis looks at a state's incentives as a ratio to value added of the firms receiving those incentives. States with a lower prime-age employment rate tend to have higher incentives, although as shown by the slope and the shaded area, the relationship is not strong or statistically significant. Large differences in incentives occur across nearby states for no clear economic reason. Compared to Illinois, Indiana's incentives are twice as high. As of 2015, Indiana's incentives averaged 2.68 percent of value-added for export-base industries; Illinois's were 1.35 percent, even though the two states have similar employment rates. Compared to North Carolina, South Carolina's incentives are twice as high; South Carolina had incentives of 2.39 percent, versus 0.93 percent in North Carolina, even though the two states have similar employment rates.

The ratio of state and local incentives to business value-added tripled from 1990 to 2015, as shown in Figure 2. Most of this increase occurred by 2001. (The jump from 2000 to 2001 was due to New York expanding its Empire Zone program statewide.) From 2001 to 2015, incentives were stable on average. Some high-incentive states, such as New York and Michigan, made cuts. Some low-incentive states, such as Wisconsin, made expansions.

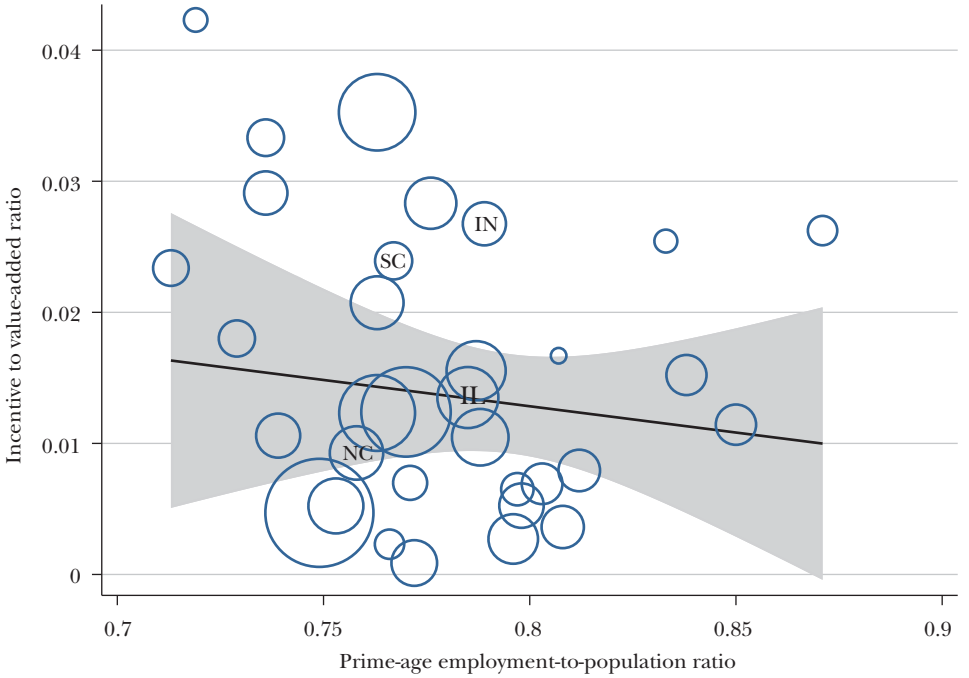
In sum, the current place-based jobs policies of state and local governments emphasize cash incentives to large corporations, without much targeting across industries or geographic areas. These patterns suggest some reforms that are likely to be desirable.

The Welfare Economics of Place-Based Jobs Policies: Some Preliminary Considerations

In the following four sections, I consider why policymakers might want to subsidize job creation in particular local labor markets, for either distributional or efficiency reasons. But first, I highlight special characteristics of local labor markets that are most relevant to this welfare economics analysis. These special characteristics include imperfect mobility and local job multipliers. I also highlight the issue of whether this policy intervention can be adequately addressed by state and local governments, or whether there is a need for federal involvement.

If mobility across local labor markets was perfect, the welfare economics analysis of place-based jobs policies would be simple, but uninteresting. Any benefits of local job creation would be fully capitalized into higher land values.

Figure 1
Incentive to Value-Added Ratio versus Employment Rate, Compared across States, as of 2015

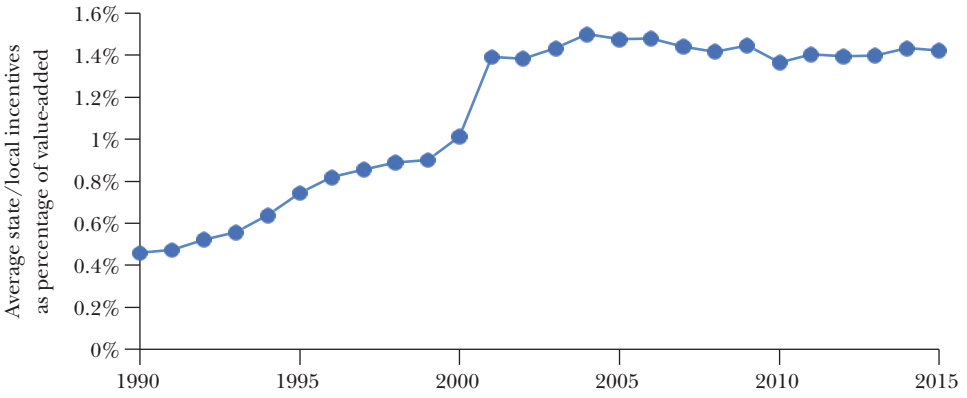


Note: Each circle represents one state. Vertical axis shows each state's 2015 ratio of the present value of incentives to the present value of value-added, for export-base industries as of 2015 from Bartik (2017b). Horizontal axis shows prime-age employment rate (ages 25–54) for each state as of 2015 and is from US Bureau of Labor Statistics (2015). Regression line and 95% confidence interval is shown based on weighted regression using 2015 state population (ages 25-54) as weights. Observations are on the 33 states with incentives data in Bartik (2017a). Circle size corresponds to state's 2015 population (ages 25-54). Four states referred to in text are initialed.

But people are hard to move. As Adam Smith (1776, Book I, Ch. 8) wrote, "a man is of sorts of luggage the most difficult to be transported." In cross-sectional data, about half of all Americans live within 30 miles of their birthplace (Zabek 2019). In panel data, about 55 percent of all Americans spend most of their career in their childhood metropolitan area (Bartik 2009). For college-educated workers, 40 percent spend their career in their childhood metro area. Percentages staying are not much lower in metro areas that are smaller or slower-growing.

These staying percentages reflect enhanced life satisfaction that people gain from the familiar people and places of their home area. As Roger Bolton (1992, p. 193) wrote, "The sense of place is an intangible, location-specific asset; it is capital . . . [People's] appreciation of [a sense of place] is evidenced by the one bit of evidence that ought to make economists notice: people are willing to pay for it." Estimates of the moving subsidy required for the median person to be indifferent to

Figure 2
Average State/Local Incentives as Percentage of Value-Added, 1990–2015



Source: From Bartik (2017a).

their current location and an otherwise similar location often exceed 100 percent of annual income (for a few recent examples, see A. Bartik 2018; Balgova 2018; Gregory 2017; T. Bartik 2019b includes further references). This required moving subsidy is 43 percent greater if a person's family is nearby (Kosar, Ransom, and van der Klauuw 2020).

But some people do move across local labor markets. Annual migration rates across metro areas exceed 2.5 percent, and five-year migration rates across metro areas or commuting zones exceed 10 percent (Molloy, Smith, and Wozniak 2011). For the thesis of this article, an important issue is the extent to which mobility leads the benefits of local job creation to be capitalized into higher land values rather than increasing local employment rates.

流动性在多大程度上导致地方就业新增资本化为土地价值的上升而非地方就业率的提高

Local job creation policies should be cognizant of local job multipliers and how they vary across industries. If incentives are provided to non-export-base industries, the net effect on local job creation is negligible. Suppose an incentive encourages a local McDonald's to add jobs; this is likely to do little more than reduce sales and jobs at the local Burger King.

In export-base industries, local job multipliers are due to local supplier effects, worker demand effects, and agglomeration economies in high-tech industries. (High-tech agglomeration economies are discussed later.) Supplier-effect multipliers occur when a local expansion of some export-base firm leads to increased demand for local suppliers to that firm. Such supplier-effect multipliers will be higher for industries which use denser networks of local suppliers, like the auto industry. Worker-demand multipliers occur when workers in the expanding export-base firm, or its local suppliers, buy more locally produced or distributed retail products. Such worker-demand multipliers will be higher when worker wages are higher in the expanding export-base firm or its suppliers.

The local supplier effects and worker demand effects do not reflect any real externalities that would lead to market failures. These external effects on local suppliers and retailers are mediated through the market, via elastic responses to small changes in local prices facing suppliers and retailers. But if local job creation is sub-optimal for other reasons, to be discussed later, then the magnitude of local job multipliers determines how much net local job creation will occur if we induce job creation in an export-base firm. Higher local job multipliers increase the benefit-cost ratio for place-based jobs policies.

Average local job multipliers range from 1.3 to 1.7 (Bartik and Sotherland 2019). In other words, for every ten local jobs created in an export-base firm, another three to seven local jobs will be created. Multipliers will be higher for high-tech industries due to agglomeration economies (as will be discussed later).

In analyzing distributional or efficiency issues in local job creation, an important question is whether the appropriate policy response is best done at the state and local level or the federal level. Which level of government should be responsible is partially a technical economics issue: if state and local governments optimally correct for market failures of particular local labor markets, are there any remaining efficiency or equity issues at the national level that might rationalize federal intervention? Which level of government should "optimize" the pattern of local job creation is also a political issue, that is, at what level of government is wise action more likely? I will return to federal versus state/local responsibility throughout this article—in particular, in the penultimate section.

Distributional Rationale for Place-Based Jobs Policies

Before considering efficiency rationales for place-based jobs policies, I briefly consider the distributional rationale. Helping create jobs in distressed places could be argued to advance equity. Local labor markets with low employment to population ratios will tend to have lower incomes per capita. Place-based jobs policies, targeted at such distressed places, could be a mechanism for redistributing income to lower-income persons.

By itself, a concern for equity does not provide a strong case for place-based jobs policies. Place-based jobs policies would seem to be dominated by policies that target lower-income persons directly, such as "making the tax system more progressive or strengthening means-tested transfer programs" (Kline and Moretti 2014, p. 633). In the absence of market failures that allow local job creation to have benefits much greater than costs, place-based jobs policies are an unattractive way to help lower-income persons. To use Arthur Okun's (1975) "leaky bucket" metaphor, if we're trying to use a bucket to give water to the poor, the bucket of place-based jobs policy could be argued to be "leakier" than other buckets that might aid the poor.

One reason this bucket might be leaky: the benefits of place-based jobs policies may be diverted to upper-income groups. If there is extensive capitalization of local job creation into land values, then place-based jobs policies would have more of their benefits go to landowners, who tend to have higher incomes. If the costs of

subsidizing business to create jobs are large per job created, then place-based jobs policies would have more of their benefits going to owners of capital, who tend to have higher incomes. A later section will consider plausible magnitudes of capitalization and costs per job based on the empirical literature.

But as we will address, market failures can lead local job creation to have benefits greater than costs. Furthermore, these local benefits are distributed modestly progressively. Therefore, place-based jobs policies can potentially be a good way to deliver benefits to lower-income persons in distressed places. Maybe the "bucket" of place-based jobs policy somehow yields more water than you originally placed in the bucket. Equity concerns could rationalize pursuing place-based jobs policies more aggressively in distressed places and with federal support for such targeting.

Place-Based Jobs Policies and Market Failures, Part 1: Involuntary Unemployment

In the next three sections, I will address the role that place-based jobs policies can play in overcoming three different types of market failures that might cause local job growth to be inefficiently low: 1) involuntary unemployment; 2) agglomeration economies; and 3) problems in markets for various business inputs. The first two categories imply that local job growth may provide benefits external to the individual firm, and because firms do not internalize these external benefits, firms on their own may underprovide jobs. The last category analyzes how local job growth is impeded by inefficiencies in markets for business inputs. For more detailed discussion of place-related market failures, useful starting points are Bartik (1990), Kline and Moretti (2014), Glaeser and Gottlieb (2008), and Austin, Glaeser, and Summers (2018).

An understanding of these market failures gives clues to how to reform place-based jobs policies to increase benefits relative to costs. In particular, an important issue is whether external benefits of new jobs are asymmetric across places. After all, job gains in one place might easily be offset by job losses elsewhere. This is obviously true for a new plant which could have located elsewhere, but it is also true for local job growth due to start-ups or expansions. Greater national market share for firms in one place will mean losses of market share of firms elsewhere. Only if external benefits from jobs are asymmetric across places can redistributing jobs across places have net national gains. Identifying where jobs have more external benefits is useful advice for policymakers. Indeed, if places with greater external benefits don't do enough to attract jobs, perhaps the federal government should encourage further job redistribution.

Involuntary unemployment makes it likely that benefits from jobs are higher in distressed places. The social benefits from a higher employment rate include both the private benefits to individuals who otherwise would not be employed (that is, higher earnings minus the opportunity costs of their non-work time[4]), and external benefits like lower crime, benefits for family members, and local fiscal benefits.

[4]As discussed in Bartik (2012), one can visualize the social benefits of new jobs by thinking of how they create a vacancy chain terminated by increases in the local employment rate or population. Social

Even in nondistressed areas, the private benefits to the otherwise nonemployed are at least 40 percent of earnings, after subtracting opportunity costs of reduced non-work time (Mas and Pallais 2017; see also Haveman and Weimer 2015). In distressed places, both the benefits to those who otherwise would not be employed and the external social benefits are likely higher. In distressed areas, an increase in local jobs boosts total earnings for the area by more, by increasing local employment rates more. Simulations presented in Bartik (2015) suggest that in an initially low-unemployment area (4 percent unemployment rate), a 1 percent jobs boost will increase the local employment rate after 10 years by 0.20 percent. But in an initially high-unemployment area (7.1 percent unemployment), a similar-sized jobs boost will increase the employment rate after 10 years by 0.34 percent. Thus, the employment rate effect in a high-unemployment area is over two-thirds greater than in the low-unemployment area, and as a first-order approximation, local earnings per capita will also go up by over two-thirds more. Similarly, Austin, Glaeser, and Summers (2018, my calculations based on their Table 4, column 2) find that the impact of local job growth on prime-age male employment rates is 61 percent greater in areas with high nonemployment compared to areas with low nonemployment.

Why do local employment rates respond more to a job shock in distressed places? Consider what happens when a firm hires for new local jobs. The firm's hires come from three sources: 1) the local employed; 2) the local nonemployed; 3) in-migrants. Any hires from the local employed leads to local job vacancies, which will be filled in the same three ways. This job vacancy chain is only terminated when the original local jobs created lead to either additional jobs for the local nonemployed or for in-migrants. This makes sense because local employment is the product of the local employment rate and the local population, and thus a log percentage shock to local employment must equal the sum of the log percentage change in the local employment rate plus the log percentage change in the local population. All along this job vacancy chain, firms face choices about whether to hire the local nonemployed or in-migrants. Higher local nonemployment rates do mean that more local labor is readily available, as revealed by the empirical finding that employment rate effects of job shocks increase with greater local distress, which reflects changes in firms' decisions about who to hire.

Overall, job growth in more distressed areas will have greater private benefits for local workers, both because more local workers will obtain jobs in distressed areas and because the local workers who get jobs will value the jobs more relative to their opportunity costs. Furthermore, social benefits of job growth are also likely to be greater in distressed areas. Larger effects on local employment rates are likely to lead to greater crime reductions and greater improvements in outcomes for children. Fiscal benefits are also likely to be higher in distressed areas, both because of increased state and local tax revenue and because distressed areas are more likely to have excess capacity in infrastructure, which reduces needed spending for new infrastructure.

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benefits can be measured by summing gains over the chain, or by the earnings gain minus the value of foregone non-work time for the otherwise nonemployed.

A common belief is that place-based jobs policy is a "zero-sum game," as any jobs gained in one place are argued to be lost to other places. What these greater benefits of job growth in distressed areas mean, however, is that reallocating jobs to distressed areas has net national benefits. Furthermore, this job reallocation might increase total national employment. This job reallocation moves labor demand to where effective labor supply responds more elastically without excess price pressures (Austin, Glaeser, and Summers 2018; Bartik 2001, Appendix 9).

On narrow grounds of efficiency, this market failure does not necessitate federal involvement. To maximize their own local benefits, state and local governments should provide the optimal subsidy for local job creation, equal to the marginal local benefits of job creation, which will increase with the local area's economic distress. Federal intervention could be rationalized on equity grounds to help distressed areas pay for the optimal job creation subsidy. I return to this topic in discussing policy reforms.

Place-Related Market Failures, Part II: Agglomeration Economies

External benefits of local job growth also occur because of agglomeration economies, which are productivity gains associated with city size or industry clustering (for an overview of this research, see the "Symposium on Productivity Advantages of Cities" in this issue). A larger scale of a city or industry cluster allows for better matching with more specialized suppliers and workers and for more knowledge in workers and firms to diffuse among local firms.

Because of agglomeration economies, job growth in one firm may have external benefits for other local firms. These external benefits are greater in certain industries, like high tech. However, research has not reached a consensus about how such external benefits are asymmetric across places. Will one more high-tech job produce greater agglomeration economies in Silicon Valley or in Detroit? We don't have a consensus answer.

Some research suggests that productivity may go up continuously with high-tech concentration so that high-tech job growth will have greater productivity spillovers in Silicon Valley than in lesser high-tech centers (Moretti 2019). Other research finds that local high-tech job multipliers are greater in areas whose high-tech concentration exceeds the national average but do not vary much among these above-average high-tech centers: for example, Silicon Valley compared to Minneapolis-St. Paul (Bartik and Sotherland 2019). These disparate results can be reconciled by allowing for congestion effects that high-tech clusters have on local costs due to higher housing prices and wages. The high-tech job multiplier in Bartik and Sotherland (2019) reflects both productivity spillovers and congestion effects.

This argument—that high-tech is similarly productive in many places—is an underlying assumption behind recent proposals for the federal government to encourage high-tech to diversify geographically. Jonathan Gruber and Simon Johnson (2019), in *Jump-Starting America*, propose that the federal government fund 20–30 high-tech centers. In a similar spirit, Atkinson, Muro, and Whiton (2019) offer

a more modest proposal for 8–10 high-tech centers. Gruber and Johnson (2019) argue that "a national strategy of developing new tech hubs [is] a cost-effective way to take advantage of local research spillovers and agglomeration." In their view, "conducting research in the existing small set of coastal locations is unambiguously a lot more expensive than doing the same in lower-cost locations elsewhere in the country." Geographic diversity of high-tech is desirable because "while there may or may not be some economic costs to redirecting public R&D towards new locations, there are unambiguous political gains." Their political argument will be considered further below.

Thus, the consensus is that there are efficiency gains from high-tech job growth in areas with above-average high-tech concentrations. But almost all high-tech jobs go to such areas anyway—mostly to a few high-tech clusters on the coasts (Johnson and Gruber 2019; Atkinson, Muro, and Whiton 2019). We lack consensus on which of the following two strategies would be better: diversifying high-tech away from coastal high-tech centers to other high-tech centers or concentrating high-tech more in coastal cities by reducing their costs through zoning and housing code reforms (Hsieh and Moretti 2019).

沿海高科技中心较高的住房成本

Despite this lack of consensus, agglomeration economies are relevant to place-based jobs policies because of how they affect multipliers. Local high-tech job multipliers probably average at least 1.9, compared to average job multipliers of 1.3 to 1.7 (Bartik and Sotherland 2019). In the many local labor markets with above-average high-tech clusters, local high-tech job multipliers may be as great as 2.5 to 3. Some studies estimate even larger high-tech multipliers (Moretti 2010). Place-based jobs policies can be more cost effective if they target high-tech firms with higher job multipliers.

One qualification to advocacy of high-tech targeting: it is unclear whether targeting high-tech has the same benefits for boosting local employment rates. Because many tech jobs are more skills-intensive, high-tech job growth by itself may not be well-matched to the job skills of the local nonemployed. However, many of the multiplier jobs will be less skills-intensive. The net effect on local employment rates of high-tech job growth, along with its multiplier jobs, needs more research.

Is federal intervention needed because of agglomeration effects? On efficiency grounds, no. Agglomeration gives state and local governments a reason for an optimal job creation subsidy varying by industry.

Federal intervention has been argued for on distributional grounds: it seems unfair to have most high-tech growth go to only a few coastal cities. Gruber and Johnson (2019) also make a political argument: more geographically diverse high-tech growth will increase political support for expanding federal R&D spending. The distributional case can be questioned: how will redistributing high-tech jobs from Silicon Valley to Minneapolis-St. Paul help low-income persons? The candidate high-tech centers in Gruber and Johnson (2019) or in Atkinson, Muro, and Whiton (2019) mostly seem relatively prosperous already. The political case also can be questioned. Why will rural Wisconsin residents, who already feel that the state capital of Madison gets unfair advantages (Cramer 2016), be more supportive of federal research and development if Madison is picked as a high-tech center?

Place-Relevant Market Failures, Part III: Public Services and Regulation Affecting Business Inputs

Place-based jobs policies can become more cost effective by being cognizant of market failures affecting business inputs. Business input costs may be affected by local public services and regulations. Here, I will focus on three local public policies affecting business inputs: customized business services, infrastructure, and land development. Suppose that per dollar spent on such programs, we can reduce business input costs by more than a dollar. In that case, such changes in public services and regulations can be more cost effective in promoting local job growth than tax incentives and other cash incentives. After all, providing cash to a business through tax incentives, at best, lowers business costs by one dollar per one dollar of government costs. These inefficiencies in business input provision do not necessitate federal intervention because state and local governments should be interested in creating jobs at lower costs.

Customized Business Services

Customized business services include manufacturing extension services for small manufacturers, customized job training provided by community colleges, small business development centers, and business incubators providing space, networking, and advice.

As one example, we already have a network of manufacturing extension services across the country funded by government and some private fees. These extension services provide advice to smaller manufacturers on adopting new technology, finding new markets, and other issues. Sometimes the advice, particularly if it is short-term, is directly provided for free by the extension office. Other times the advice is provided on a longer-term basis at a fee or by a network of high-quality advisors screened and recommended by the extension office. Some state extension services have links to local universities and community colleges and will provide referrals and some subsidies for manufacturers to receive consulting services from faculty.

Customized business services can be rationalized by several market failures. Many small firms lack adequate information, and these firms may also find it difficult to finance such services because of imperfect capital markets. The quality of information is hard to evaluate, and marginal costs of providing such information are low. Public agencies can provide higher-quality information or certify the quality of private or public consultants.

Customized job training includes both general and firm-specific training, which cannot be easily separated. Firms may underinvest in training if they fear worker exit. Such fears may be more acute for small firms. Smaller firms may also lack the scale to run training efficiently.

Quasi-experimental studies suggest productivity benefits for firms of at least five times the public costs for both manufacturing extension (Jarmin 1999) and customized job training (Holzer et al. 1993). The manufacturing extension evidence compares similar firms whose likelihood of receiving services varied with proximity

to the extension office. The customized training evidence compares similar manufacturers that applied at various times for a program that awarded grants on a first-come, first-serve basis. More recent survey evidence backs the effectiveness of both manufacturing extension (Robey et al. 2018) and customized job training (Hollenbeck 2008).

Public Infrastructure

In some cases, infrastructure can be a cost-effective way of promoting job growth in depressed regions. In one historical example, the Tennessee Valley Authority (TVA) emphasized infrastructure such as dams, highways, canals, and electrification. In research that compares the TVA region with similar regions that were unsuccessfully proposed for similar assistance, TVA is found to boost the region's manufacturing jobs by over 250,000 (Kline and Moretti 2013). Similarly, the Appalachian Regional Commission (ARC) focused two-thirds of its funding on highways. ARC highways had cumulative effects of increasing local jobs by 5.2 percent and increasing per capita incomes by 1.3 percent (Jaworski and Kitchens 2019). The present value of ARC highway investment is $85.1 billion, and the 1.3 percent boost in annual incomes is $13.4 billion.

Land Development

Land development in the United States is regulated by zoning, building codes, and tort litigation. Providing land that is appropriately zoned is widely viewed as having major effects on local job growth. Greater availability of industrially-zoned land affects development (Chapple 2014). Local economic developers believe making more land available is an effective development strategy. Promoting business development through "industrial parks" is a regular part of the local economic developer's tool kit.

Key Empirical Estimates Relevant to Place-Based Jobs Policies

Based on the above discussion, the strongest efficiency and equity rationale for place-based jobs policies is for creating jobs in distressed places. But is this rationale backed by estimates of benefits versus costs? Or are estimated benefits too small or too capitalized into land values? Or are estimated job-creation costs too high?

Before considering empirical estimates relevant to local job-creation's benefits and costs, I consider an oft-proposed alternative way of helping people in distressed areas: why not move people to jobs? Can a strategy to encourage out-migration from distressed areas work?

Moving People to Jobs Is an Ineffective Strategy 人随产走

Encouraging out-migration from distressed places has two problems: 1) it's hard to get people to move; 2) out-migration does not help those left behind.

Even though annual and five-year migration rates among metro areas are large, getting additional people to move out of a distressed local labor market is costly.

Suppose we offer sizable subsidies for moving out of a distressed area, like a subsidy of $10,000. Subsidies in this range would increase out-migration rates by about 2 percentage points (Kennan and Walker 2011; A. Bartik 2018).

Even persistent local job loss has small out-migration effects. Autor, Dorn, and Hanson (2013, pp. 2141–42) "find no robust evidence that [Chinese trade–induced] shocks to local manufacturing lead to substantial changes in population." Going from the 25th to the 75th percentile of exposure to trade with China across commuting zones increases outmigration rates over a 10- to 14-year period by less than 1 percentage point, despite substantially lowering earnings (A. Bartik 2018).

Encouraging out-migration from distressed areas has another problem: it does little to re-equilibrate labor demand and supply in the distressed area. Migration shocks to local labor supply induce shocks to local labor demand of similar or larger size: for example, Beaudry, Green, and Sand (2018) and Howard (forthcoming) both find an elasticity of local jobs with respect to local population shocks of 1 or slightly higher; T. Bartik (2019b) provides a further review of the relevant literature. For every X percent of working-age people that leave Flint, Michigan, the jobs in Flint will go down by an amount similar to X percent, or perhaps more.

Why does migration affect local jobs? Migrants bring assets and transfer income, which affect local demand. Migrants demand housing, affecting construction. Migrants may be entrepreneurial, affecting business start-ups. Migration affects property values, which affect the local consumption of property owners (Howard forthcoming). Out-migration reverses all of these patterns.

A policy analysis of subsidizing out-migration from distressed communities should include all benefits and costs. Perhaps the out-migrant will get higher earnings. But if out-migrants choose not to move without the subsidy, their lost sense of place in their community may exceed the earnings boost they receive. If a person's unemployment imposes costs on their family, perhaps the family benefits of the move outweigh the individual's loss. But moving puts stress on children, which leads to behavioral problems, substance abuse, and poor mental health (Coley and Kull 2016; Oishi 2010). Subsidizing out-migration from distressed communities may often create as many problems as it solves.

In short, the problems caused by low employment rates in local labor markets cannot be substantially solved by encouraging people to leave for distant jobs. The main alternative is bringing jobs to people, or place-based jobs policies, which I turn to next. Do plausible empirical benefits of place-based jobs policies exceed plausible job-creation costs?

Benefits from Local Job-Creation: Is Everything Capitalized in the Long-Run?

Some economists have raised concerns that in the long run, benefits from job growth will be largely capitalized into land values (Glaeser and Gottlieb 2008; Marston 1985; Winnick 1966). In this view, higher employment rates and wages attract migrants, which will raise property values and push employment rates and real wages toward their prior equilibrium. Indeed, if capitalization were to dominate, local job growth would ultimately have regressive effects as property ownership is

concentrated in upper-income groups. But capitalization does not dominate; local job growth has sufficiently large benefits for local labor market outcomes, both in the short-run and more importantly in the long run.

An increase in local jobs has large short-run effects on employment rates: a short-run elasticity of about 0.6 (for reviews of 18 US studies, see T. Bartik 1993, 2015). Perhaps more surprising is that a one-time boost to the level of local jobs has large effects on local employment rates after 10 or more years. The consensus from the studies listed in Table 2 is a long-run elasticity of between 0.2 and 0.3.[5] These long-run employment-rate effects of a local job shock are qualitatively consistent with other research. More severe local recessions depress local employment rates for at least 25 years (Greenstone and Looney 2010) and real earnings per capita for at least 20 years (Stuart 2017). These long-run persistent effects of more severe local recessions hold in every recession from 1973 through the Great Recession (Hershbein and Stuart 2020).

Some historical examples reinforce these long-term effects. In areas where the government built a World War II manufacturing plant, manufacturing wages are higher even 50 years later (Garin 2018). Mississippi's "Balance Agriculture with Industry" program, a pioneering incentive program begun in 1936 (Freedman 2017), attracted northern manufacturing plants, mainly textile plants with a female workforce, by offering incentives of free land and buildings and property tax breaks. The BAWI program increased female labor force participation rates in the affected counties for at least 24 years until 1960. BAWI may also have increased male labor force participation. Effects persisted after most of the original plants had closed.

Human capital is a likely mechanism for these long-run effects. In the short run, local job growth increases residents' employment rates. This job experience improves their job skills, reduces crime and substance abuse, and increases self-confidence. Greater employment rates change social norms about work, and in some cases about women working (Freedman 2017). Even after migration has fully adjusted, residents' higher human capital allows higher employment rates.

Persistent local labor market effects are consistent with research on how labor demand shocks affect individuals. Worker displacement from jobs persistently lowers earnings (Davis and von Wachter 2011). Young workers entering the labor market during a recession suffer a long-run earnings penalty (Kahn 2010; Schwandt and von Wachter 2019). Locally severe recessions reduce residents' employment rates, even if the individual moves elsewhere (Yagan 2019).

These persistent employment rate effects of job growth imply large benefits per local job created. The long-run elasticity of 0.2 for the employment rate means that each job created raises earnings persistently by about 20 percent of average earnings per job. The present value of benefits per job will be many multiples of average earnings per job. Suppose all jobs paid $60,000 annually. Assume a fixed

[5] An exception is Blanchard and Katz (1992). However, their finding of zero long-run employment rate effects is not robust to alternative estimation approaches (T. Bartik 1993, 2015; Rowthorn and Glyn 2006). Amior and Manning (2018) argue that because data limitations forced Blanchard and Katz to include only two lags in all variables, long-run responses may be biased.

Table 2

Long-Run Elasticities of Local Employment to Population Ratio with Respect to Once and For All Local Employment Shock, US Studies

Study	*Nature of estimate*	*Long-run*	*Qualifications*	*Elasticity*
Bartik (2015)	Dynamic model, panel data on MSAs at annual frequency, 1979–2011	10-years	at 4.0% unemployment rate (UR)	0.20
		10-years	at 7.1% UR	0.34*
		10-years	at 10% UR	0.47*
Bartik (1991)	Reduced form regression of change in labor market outcomes on current and lagged annual job growth, annual panel data on MSAs, 1972–1986	8 years	OLS	0.23*
		8 years	2SLS using demand shock instruments	0.37*
Blanchard and Katz (1992)	Dynamic model with two lags, panel data on states with annual frequency, 1978–1990	8 years		0.07*
		17 years		0
Bartik (1993)	Same data as Blanchard and Katz but with lags in growth added	8 years		0.28*
		17 years		0.25*
Bound & Holzer (2000)	Decade change, 1980–1990, MSAs	10 years	High-school or less College or more	0.24* 0.12*
Partridge and Rickman (2006)	Dynamic model, annual panel data on states, 1970–1998	10 years	Preferred estimates	0.21*
			Alternative estimates	0.42*
Notowidigdo (forthcoming)	Decade changes, 1980–2000, panel data on MSAs	10 years	Mean effect	0.14*
Beaudry, Green, and Sand (2014)	"Decade" changes, 1970–2007, panel data on MSAs	10 years		0.24*
Amior and Manning (2018)	Decade changes, 1950–2010, panel data on commuting zones	10 years		0.30*

Notes: Elasticity is long-run elasticity of local employment to population ratio with respect to once-and-for-all shock to local job level. Asterisk indicates estimate is significantly different from zero at 5% significance level. Bartik (2015) figures extend that paper's results to slightly different low and middle unemployment rates, set at 10th and 90th percentile of local unemployment rates from 2016 ACS. Bartik (1991) OLS results from his Figures 4.2 and 4.3; 2SLS from his Table 4A4.2. Blanchard and Katz (1992) results are for a fixed shock to job level, which makes estimates comparable to other studies (T. Bartik, 1993, Table 2, Row 2). Bartik (1993) results are from his Table 2, Row 3, and add growth terms to Blanchard-Katz specification to optimize Akaike Information Criterion. Bound and Holzer (2000) results are IV results in their Table 3 for annual per capita hours worked. Partridge and Rickman (2006) comes from text discussion of their Figure 1, which implies statistical significance. Notowidigdo (forthcoming) results from his Table 2, column 3, using mean of predicted employment (obtained courtesy of the author), and then dividing resulting effect on the employment to population ratio in points by the mean ratio in his Table 1. Beaudry, Green, and Sand (2014) is from their Table 6, column 3. They use 2000 to 2007 as one of their "decade" changes. Amior and Manning (2018) results from their Table 2, column 4.

0.2 elasticity. Then each local job created increases employment rates sufficiently to increase earnings per capita by $12,000 (0.2 times $60,000). At a 3 percent discount rate, this earnings increase has a present value of $400,000. At a 5 percent discount rate, the present value is $240,000.

This calculation assumes that any opportunity costs of reduced non-work time, due to higher employment rates, are at least offset by social benefits like lower crime and substance abuse and higher fiscal benefits. This rough calculation also ignores

the higher short-run elasticity, as well as the possibility that this long-run elasticity might eventually depreciate. But more refined calculations are also likely to yield benefits per local job created in the hundreds of thousands of dollars.

Do Employment Rate Effects Outweigh Capitalization's Regressive Effects?

Using plausible empirical magnitudes, a demand shock to the level of local jobs affects the local income distribution progressively. The elasticity of housing prices with respect to a local job shock is 0.4–0.5 (T. Bartik 1991; see also Saiz 2010). But a shock to the local job level results in only a one-time gain for property owners. In contrast, a local job shock increases local employment rates for many years—an elasticity of 0.6 in the short run and 0.2 to 0.3 in the long run. Real wage rates also rise (T. Bartik 2015). Under plausible discount rates, the present value of these continuing increases in local per capita earnings will exceed the one-time property-value gain by a ratio of over 3 to 1 (T. Bartik 1994, 2005, 2018a).[6] Real earnings effects of local job growth are modestly progressive (Bartik 1994) in part because higher employment rates help those otherwise not employed who tend to have lower incomes.

Because earnings effects dominate property-value effects, and earnings effects are progressive, the overall effects of local job growth are progressive. Table 3 summarizes some illustrative estimates. Progressive impacts are modest. Percentage effects on the lowest-income quintile are 2.4 times percentage effects on the highest-income quintile. Because the highest-income quintile has 10 times the average income of the lowest-income quintile, however, the dollar effects of local job growth on incomes are higher for higher-income groups.

As a result, how we finance local job growth makes a big difference to whether such policies have a progressive effect. If the financing is by cutting highly progressive public spending, such as welfare (Bartik 1994) or public schools (T. Bartik 2018a), the net impact is likely to be regressive. Also, the progressivity of local job growth may vary with policy. If a firm receiving incentives hires more of the local non-employed, the employment rate effect of the new jobs will be greater. A better local workforce system might encourage hiring of the local non-employed. Capitalization is greater in local areas with more housing supply restrictions (Saiz 2010). However, even in areas with the most-restricted housing supply, the long-run earnings effects of local job growth have a greater present value than the increased housing values (T. Bartik 2018a). The match of the jobs created with local skills may also matter, but there is little empirical evidence on this point: for some simulations, see Persky, Felsenstein, and Carlson (2004).

Costs of Creating Local Jobs: High for Cash Incentives, Lower for Policies Affecting Business Inputs

[6]Why is the ratio of earnings effects to property value effects so high? In T. Bartik (2018a), locally-owned property is 3.5 times earnings, and the property value elasticity is 0.45, so the property value effect is equivalent to a one-year-only earnings elasticity of 1.6. If the earnings elasticity starts at 0.6 and declines to 0.3, and real wages also increase, it is unsurprising that labor market benefits exceed property value effects.

Even if local job creation can have high benefits per job created, do these benefits outweigh the costs of job creation? The answer is: maybe for cash incentives but more clearly for policies that enhance business inputs.

Cash incentives to encourage local job creation have high costs per job created because it takes a lot of cash to tip a business location or expansion decision. The empirical literature suggests that a cash incentive with a present value of 1 percent of value-added—slightly below the average for state and local incentives—will tip, at most, 10 percent of the location or expansion decisions of businesses awarded such incentives (for a review of the relevant literature, see Bartik 2020). At least 90 percent of the time, the business receiving the incentive would have made the same location or expansion decision, even without the incentive.

Why such modest effects? A cash incentive with a present value of 1 percent of value-added is equivalent to a perpetual 2.2 percent wage subsidy. Differences across locations in other costs—wages, labor productivity, real estate, proximity to markets or suppliers—will often outweigh a 2 percent wage subsidy. An anecdote: One CEO told me that his decision about where to locate a particular new facility had been determined by the availability in that city of an empty factory. The empty factory allowed the new facility to get into production quickly. The incentive his business received had no effect on the location decision.

Such modest effects of incentives imply that the cost per job created will be high: at least $100,000 and probably more. In 2019 dollars, value-added per full-time-equivalent worker in export-base industries is $177,000 (Bartik 2017, Table 3). According to Poterba and Summers (1995), firms use a 12 percent real discount rate in making investment decisions. At a 12 percent real discount rate, the present value of value-added per full-time-equivalent in export-base industries is $1.65 million ($177,000/0.12). Suppose an upfront incentive is provided. Based on the research, an incentive of 1 percent of the present value of value-added will tip 10 percent of such location decisions. This implies a cost per job created in incented firms of $165,000 ($1.65 million times 0.01/0.10). If the local job multiplier is 1.5, then the cost per job created, including multiplier jobs, will be around $110,000 ($165,000/1.5).

But a totally upfront incentive is not the typical incentive package. The typical incentive package continues at a high level for at least 10 years (Bartik 2017, Table 22). States prefer drawn-out incentives to up-front incentives for two reasons. First, up-front incentives pose a greater problem of clawing back the incentive if the jobs go away. Second, governors prefer to defer incentive costs to their successors. More drawn-out incentive payments will have decreased cost-effectiveness relative to their social costs as long as the social discount rate is lower than the discount rate used by firms in making location decisions. Using the typical drawn-out incentive structure in the U.S., a social discount rate of 3 percent, and a multiplier of 1.5, the model described in T. Bartik (2018a) finds a cost per job created of $196,000.

As mentioned, the benefits per job created may also be in the hundreds of thousands of dollars. Therefore, even costly incentives may have benefits greater than costs. But often the benefit-cost ratio for incentives will not be much greater than one. For example, one study finds that when incentives are targeted at

Table 3

Percentage Effects of a 1% Local Job Shock on Income Due to Earnings Effects and Property Value Effects

Type of effect	*Average percent effects on income for:*		
	All households	*Lowest income quintile*	*Highest-income quintile*
Earnings effects	0.18%	0.41%	0.11%
Property value effects	0.05%	0.03%	0.07%
Sum of earnings effects and property value effects	0.23%	0.44%	0.18%

Note: Based on simulation model of T. Bartik (2018a). This model assumes short-run and long-run employment rate elasticities of 0.6 and 0.25, and housing price elasticity of 0.45. The distribution of earnings effects by quintile is from Bartik (1994). This version of the T. Bartik (2018a) model assumes fixed 1% once and for all shock to local job level that occurs at no incentive cost. Effects calculated over 20 years and use a 3% social discount rate. Present value of earnings effects and property value effects, after state/local taxes, are calculated as percentage of present value of average market plus transfer income of each group. Based on Congressional Budget Office (2016), the baseline income shares of the lowest- and highest-income quintiles are 5.1% and 52.0%.

non-distressed areas and have an average multiplier, the estimated benefit-cost ratio is 1.5 (Bartik 2019a). This leaves little room for plausible alternative assumptions.

Incentives are more likely to pay off if multipliers are higher or if incentives are targeted at distressed areas. Higher multipliers reduce costs per total local job created. Targeting at distressed areas increases benefits per job created.

Costs per job created can be much lower for policies affecting business inputs. As discussed, some studies suggest that customized public services to business can lower business costs by about five times their costs to the government. If so, such services can have a cost per job created that is much lower than cash incentives. Using the model in T. Bartik (2018a), these customized services have a cost per job created of around $34,000. Based on surveys of firms, manufacturing extension and customized job training both have a cost per job created of under $15,000 (Ehlen 2001; Hollenbeck 2008).

Costs per job created for infrastructure programs are also lower than for cash incentives. Consider the Tennessee Valley Authority, which improved local electrification as well as providing other services. TVA cost a little around $30 billion (in 2019 dollars). As mentioned, TVA is estimated to increase manufacturing jobs by a little over 250,000 (Kline and Moretti 2013). In this large multi-state area, the multiplier effect of this job creation would be higher, plausibly at least 2 (Bartik and Sotherland 2019), so the total job creation would be 500,000. The cost per job would then be around $60,000. A more refined calculation, which corrects for the timing of TVA spending versus job creation, finds that TVA's present value cost per job created is around $77,000 (for more details, see the Data Repository for this article).

A lower cost per job created also can be achieved by making more land available for business development. For example, case studies suggest that cleaning up "brownfields"—older industrial sites with environmental contamination—may have a cost per job created of $13,000 (Paull 2008).

All these estimates have the inherent uncertainty of any research finding. But another source of uncertainty is that effects of public services to business, infrastructure, and land development practices will probably vary with both program quality and local context. The cost-effectiveness of manufacturing extension and customized job training depend on the quality of the local programs as well as on the characteristics of local businesses. The success of infrastructure investments in rural electrification by the Tennessee Valley Authority does not mean that the proverbial "bridge to nowhere" is a good idea. Whether a brownfield project or industrial park succeeds in attracting industry depends very much on "location, location, location"—as well as on the local economy.

Improving Place-Based Jobs Policies

Based on this analysis and the empirical literature, how might place-based jobs policy be improved? Here I list six possible reforms. At the outset, I emphasize that these reforms include evaluation because we need to know whether a particular program is working in a particular place.

First, place-based jobs policies should be more geographically targeted to distressed places. The benefits of more jobs are at least 60 percent greater in distressed places than in booming places. But our current incentive system does not significantly favor distressed places.

Second, place-based jobs policies should be more targeted at high-multiplier industries, such as high-tech industries. Governors may claim they want to build the future economy, but state and local governments in practice do not target high-tech. One caveat: high-tech targeting should consider how to increase the access of current residents to these jobs. One model is Virginia's recent offer for Amazon's "Headquarters II." Virginia's offer included a new Virginia Tech campus in northern Virginia and increased funding at state colleges for tech-related programs. These education programs increased the odds that Virginia residents would fill Amazon's jobs.

Third, incentives should not disproportionately favor large firms, especially given the renewed concern in economics over excess market power in product markets and labor markets (Azar, Marinescu, and Steinbuam 2017; Gutiérrez and Phillippon 2017).

Fourth, place-based jobs policies should put more emphasis on enhancing business inputs. Customized business services, infrastructure, and land development services have the potential to be more cost effective than incentives as ways to increase local jobs and earnings.

Fifth, place-based policies should be a coordinated package of policies attuned to local conditions. One area may need more infrastructure; another, training; and still another, better land development processes. Place-based policies are complementary. If the local nonemployed are more skilled, job growth increases employment rates more. If more jobs are available, it is easier to design effective training programs. Business inputs are complementary—boosting infrastructure helps growth more if the local economy also has customized business services.

Sixth, place-based jobs policies should be evaluated better. Random assignment of firms or areas receiving assistance is hard to implement. However, better evaluation is attainable by awarding assistance using quantifiable selection criteria because this method allows for regression discontinuity techniques comparing firms or areas that just made or missed the cutoff for receiving services. For example, support for small business services can be evaluated better if some quantitative scoring system is used to decide which firms receive services. Area strategies can be evaluated better if higher units of government select distressed areas using a quantitative cutoff. Some improvements have been made in transparency and evaluation for incentives (Pew Charitable Trusts 2017).

Taking these themes together, it seems plausible that an appropriate scale of place-based jobs policies could have lower costs than current policies. For a back-of-the-envelope estimate of the needed number of jobs, if we take all commuting zones in the bottom quintile of employment rates and ask how many jobs would be needed to bring their employment rates to the median, assuming an elasticity of 0.25 for the employment rate, the needed job creation is 6 million jobs. Based on the previous discussion, a job-creation strategy that focused on enhancing business inputs might create jobs at $50,000 per job. The total cost to generate or reallocate 6 million jobs to distressed areas would then be $300 billion. If pursued over a 10-year period, the cost would be $30 billion annually. This is half of current annual costs for place-based jobs policies of $60 billion. This less costly strategy would do more to enhance both efficiency and equity by bringing up the bottom quintile of commuting zones to the national median employment rate.

Is Federal Intervention Needed?

An efficient place-based jobs program is feasible without federal intervention. State and local governments should find it in their residents' interests to pursue job creation more aggressively in distressed places. All places can improve cost-effectiveness by better targeting of incentives and by diverting resources from incentives to enhancing business inputs. But the ideal isn't happening. The political temptation is strong both in distressed and nondistressed places for state and local governments to devote lots of money to incentives to large firms, without much industry targeting. Voters are more likely to support incumbent governors who offer high-profile incentives (Jensen and Malesky 2018). Should the federal government intervene?

Any intervention should be realistic about federal capacities and local needs. First, the federal government already has a lot on its plate. Federal knowledge is limited about what works best to advance economic development in particular places.

Second, any meaningful system of federalism must allow state and local governments considerable freedom to pursue their own goals for economic development in their own ways. Given the diversity of local needs and our still-evolving

knowledge about what works in economic development, there are advantages to local experimentation.

Therefore, federal intervention in place-based jobs policies should be limited to promoting clear national interests. One national interest is limiting unfair advantages for larger firms. Large incentives for the largest companies go against the national interest in limiting large firms' market power. Another national interest is helping people in distressed places. As mentioned, distressed places have good reason, on their own, to aggressively promote job creation using their own resources. But distressed places' resources are more limited. The national interest in equity would be advanced by helping pay for job-creation programs in distressed places.

How could the federal government promote these national interests while respecting local needs? I offer below some suggestions, for both limiting incentives and helping distressed places.

Capping Incentives for Large Firms

Federal limits on incentives have been previously proposed. Incentives could be subjected to extra federal taxes (Burstein and Rolnick 1995) or penalized by reducing federal grants (Chatterji 2018; LeRoy 2012).

But such regulation of incentives raises two problems. First, there is the problem of how to define an "incentive" versus normal tax policy and how to administer any federal restrictions, given the thousands of incentives handed out each year. Second, such incentive regulation might imply federal micro-management of the state and local business tax system and state and local economic development policy.

To overcome these problems, the federal government could pursue a more modest goal: capping incentives to the largest firms. The federal government could enforce such limits by an outright ban, an extra tax, or withholding federal grants. What would be limited is discretionary incentives to firms with more than, say, 10,000 employees that exceeded some percentage of the investment or annual payroll in the new jobs created. States would remain free to provide incentives to all firms, all export-base firms, or all manufacturing firms. States could decide how much they want job growth, and in large measure how best to pursue it. What would be restricted is providing extra incentives to large firms simply because they are already large.

Targeting incentive restrictions on larger firms reduces the federal government's administrative challenges in managing incentive restrictions. As of 2016, only 1,491 US firms had more than 10,000 employees (Longitudinal Business Database, US Census Bureau 2018). Such firms are 28 percent of business employment but receive over half of state and local incentives. Restricting incentives to such large firms might reduce incentives by one-third.

This proposed incentive restriction has a precedent in European Union rules on regional state aid (LeRoy and Thomas 2019). The European Union limits the magnitude of such state aid, with stricter limits on larger firms. The European Union also makes the limits stricter in less-distressed areas.

Helping Distressed Areas with Flexible Block Grants

The federal government could help distressed areas by providing economic development grants. Many expanded spending programs for distressed places have been recently proposed: infrastructure spending (Center for American Progress 2018; Smith 2018); public jobs programs or nonprofit jobs programs (Center for American Progress 2018; Neumark 2018); employer subsidies for job creation or hiring, with some targeting on disadvantaged workers (Glaeser, Summers, and Austin 2018; Neumark 2018); manufacturing extension (Baron, Kantor, and Whalley 2018; Bartik 2010); customized job training (Austin, Glaeser, and Summers 2018; Bartik 2010); and help for small business or entrepreneurs (Chatterji 2018)

However, helping distressed places with only one specific program has problems. A single-program approach assumes that researchers know the "one best program." We don't. A single-program approach assumes that all distressed places need the same program. They don't. A single program does not allow for the synergy from simultaneously pursuing multiple programs. For example, a wage subsidy to encourage the hiring of disadvantaged workers will create more jobs and have smaller displacement effects if it is combined with customized business services that promote local job creation in businesses with high multipliers. A better federal approach to helping distressed areas is to provide a flexible block grant with many allowable uses, which can then be attuned to local needs.

This federal block grant program can also promote better evaluation. As noted earlier, the federal government could design its selection of distressed areas so that distressed and nondistressed areas can be compared for evaluation purposes.

Conclusion

Places matter for policy because places matter to people. Our knowledge about local labor markets should inform how we help distressed places. Our most important knowledge can be summarized in two numbers discussed earlier in this essay: 1.0 and 0.2. A shock to local population has an elasticity of about 1.0 in its effects on local jobs. Therefore, moving people from distressed to nondistressed places does not help restore local labor market equilibrium. A shock to local jobs of 1 percent increases the local employment rate in the long run by at least 0.2 percent. This allows place-based jobs policy to have large long-run benefits that are distributed progressively.

Place-based jobs policy needs additional research. As this essay has argued, reformed place-based jobs policies can have higher benefit-cost ratios than current policies. The targeting of distressed places could be improved with a more extensive research basis for defining distressed places and identifying which programs are most cost effective in different places. But the existing evidence clearly shows that adding jobs in distressed places offers both private and social benefits.

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